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Class: BCT 'A'

Q) Define Replication. What are the advantages of replication? Point out its drawbacks.

→ Replication is the process of sharing data across multiple machines (servers) to ensure consistency, reliability, ~~fault~~ fault tolerance, and availability in a distributed system.

Advantages of replication are;

i) Fault Tolerance;

If one server fails, others can continue to serve. ~~Guar~~ Guarantees strictly correct behaviour even in presence of certain number & type of faults.

ii) High Availability;

Response is obtained within a reasonable time bound. Users get uninterrupted service.

iii) Load Balancing;

Replicas can distribute read & write requests, thereby balancing the load.

iv) Scalability;

Adding more replicas can scale system horizontally.

Faster Access:

Data can be placed closer to user geographically.

Drawbacks:

- Consistency Maintenance:
Keeping all replicas updated is complex.
- Increased Storage:
More disk space is required.
- Synchronization Overhead:
Replicas must be kept in sync, increasing communication cost.
- Conflict Resolution:
Concurrent updates can lead to conflicts.

2) What are the different requirements of replication in distributed system?

→ Different requirements of replication in distributed system are:

i) Transparency:

Ensures that the complexities of replication are hidden from users and applications, simplifying system management and improving fault tolerance.

ii) Consistency:

Determines how data updates are propagated across replicas, ensuring that clients see a coherent & correct view of distributed data.

iii) Fault tolerance:

Should tolerate server crashes, network partitions, etc. It always guarantees strictly correct behaviour even in the presence of a certain number and types of faults.

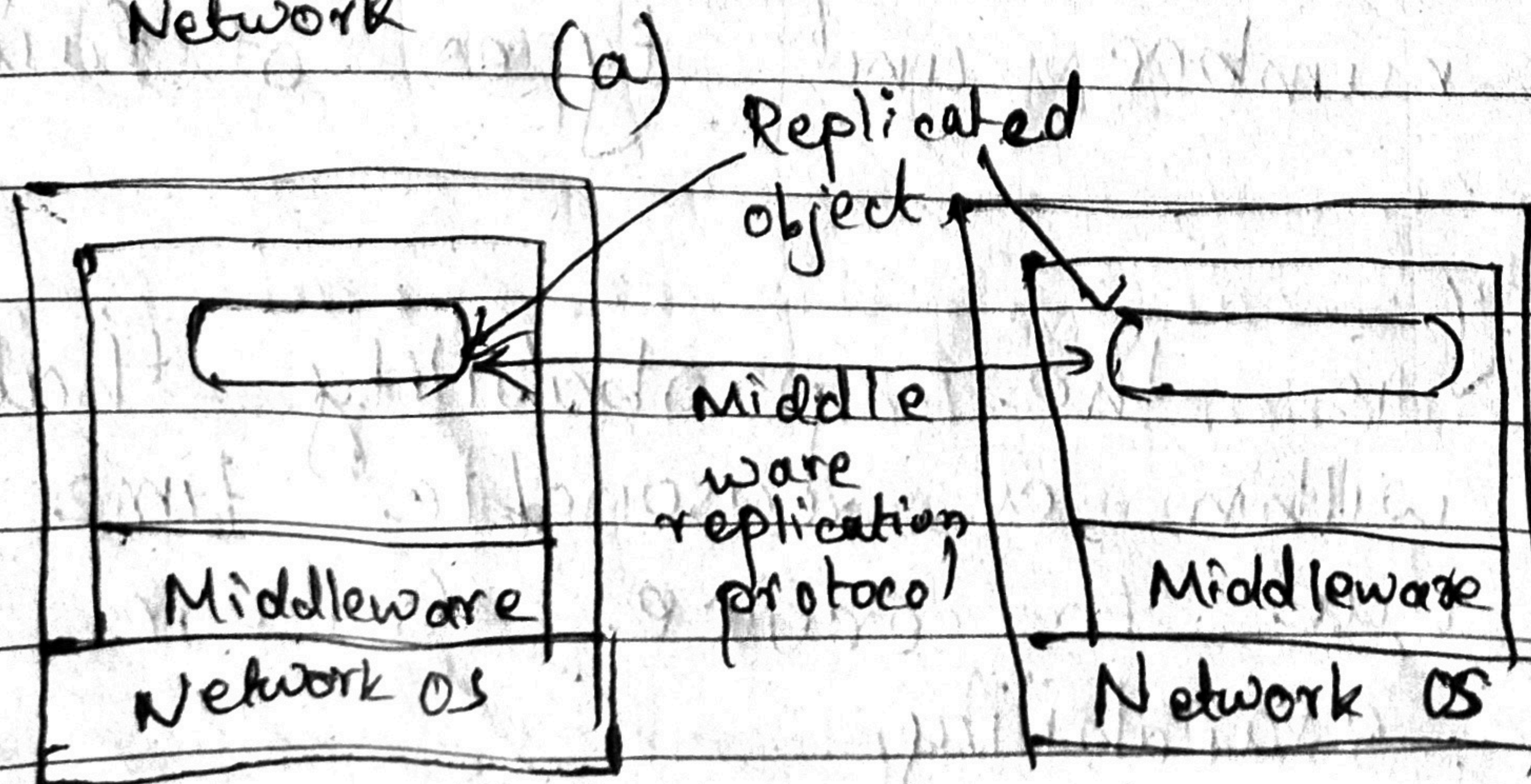
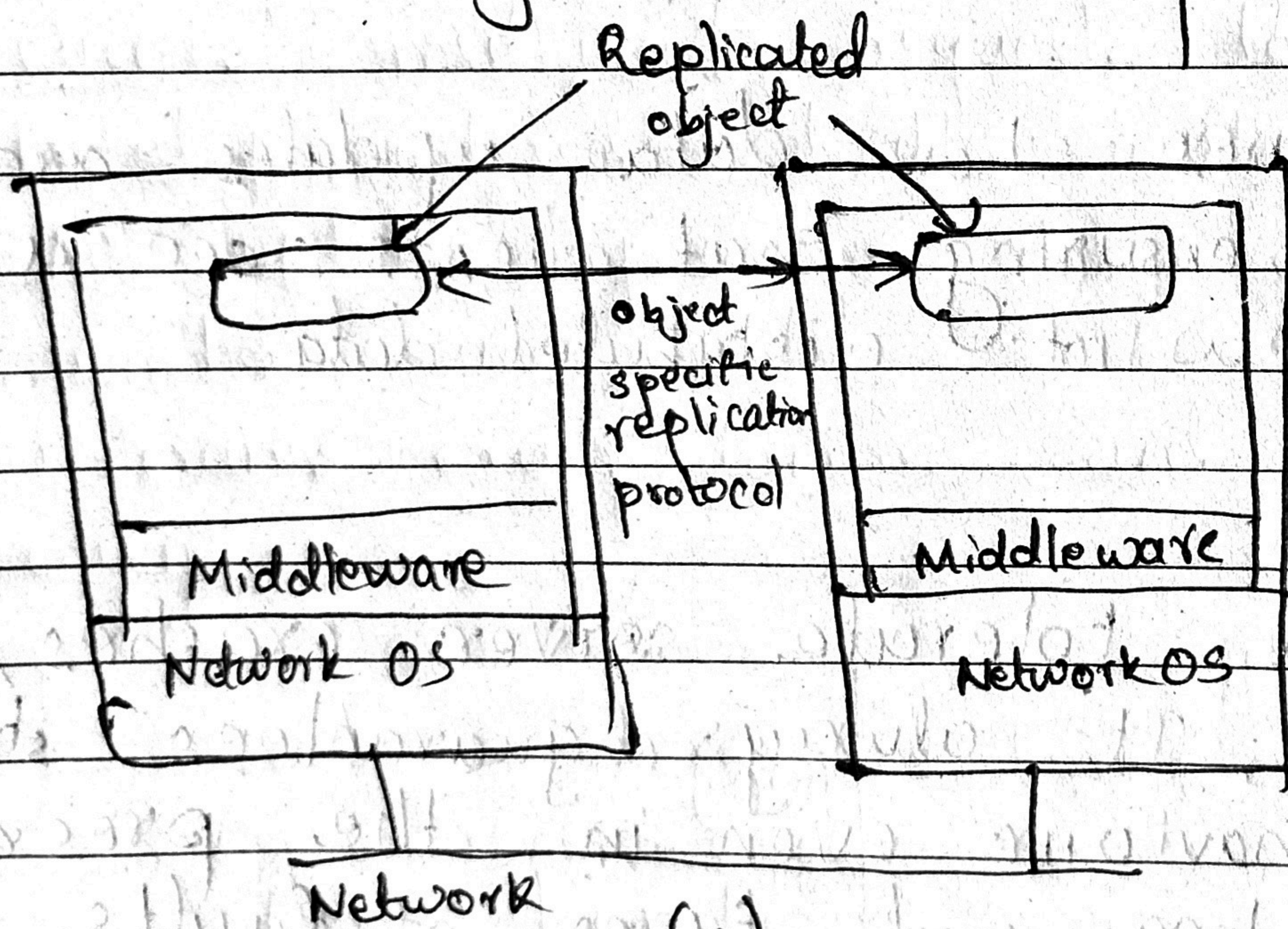
iv) Availability:

It is the probability that response is obtained within a reasonable time bound. Delays can be due to service features. Redundancy enhances availability.

1) Latency Efficiency: Replicas must respond quickly to read/write requests.

2) Define replica manager(RM). Draw & explain the replication architecture.

3) Replica Manager (RM) is component that manages access to a replica. It handles client requests (read/write), ensures data consistency, & participates in synchronization protocols with other RMs.



Two approaches for object sharing:

- The object itself can handle concurrent invocation.
- The object is completely unprotected against concurrent invocations, but the server in which the object resides is made responsible for concurrency control.

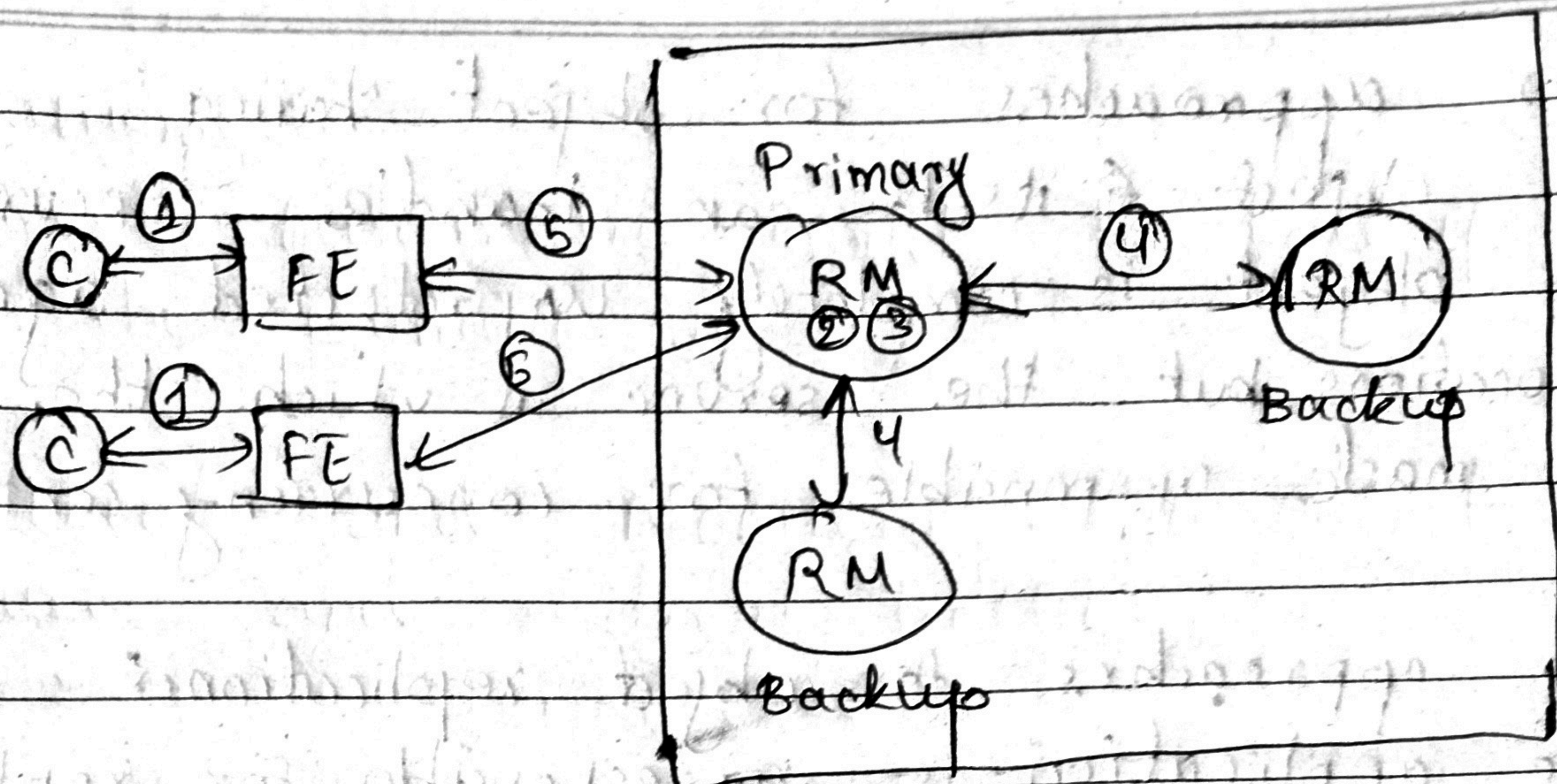
Two approaches for object replications:

- The application is responsible for replication.
 - The application needs to handle consistency issues.
- The system (middleware) handles replication.
 - Consistency issues are handled by the replication.
- It simplifies application development but makes object-specific solutions harder.

Q) Explain the different replication models with suitable examples.

→ Different replication models are:

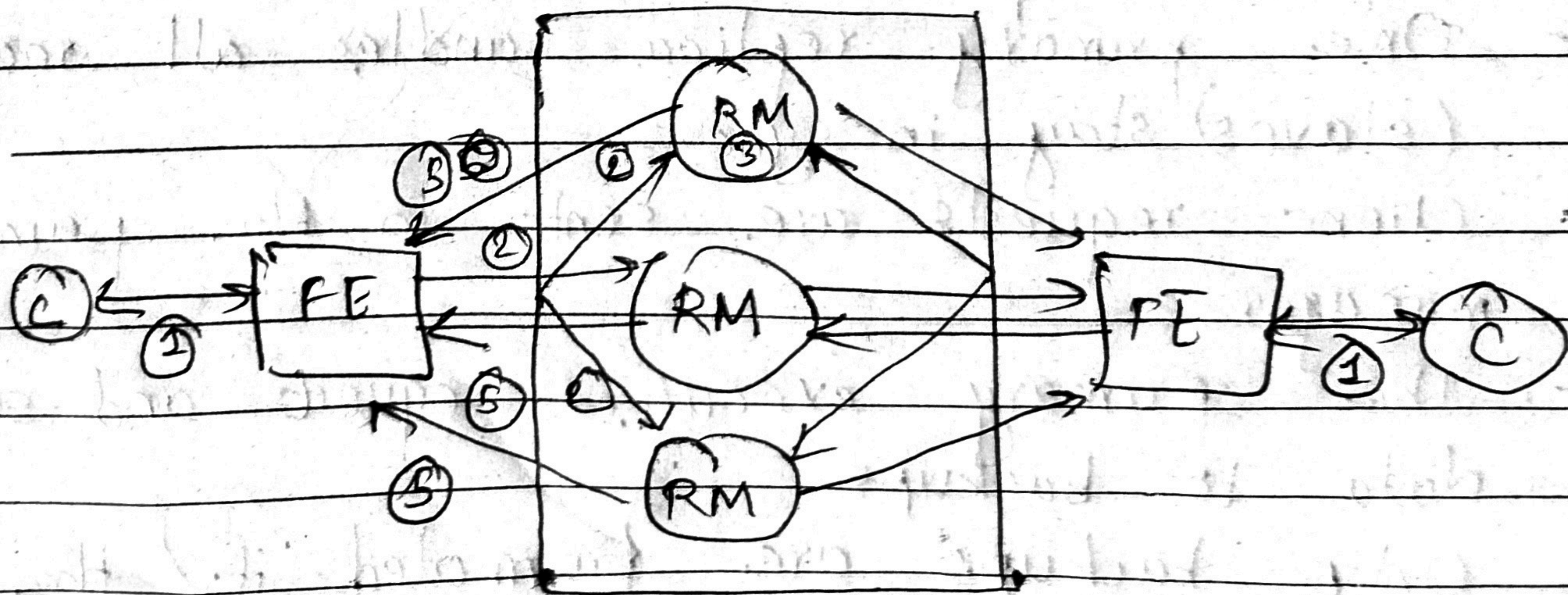
- Passive Replication
 - One primary replica handles all requests; backups (slaves) stay in sync.
 - Client requests are sent to the primary replica manager.
 - The primary executes request and sends updated data to backups.
 - Only backups are promoted if the primary fails.
 - Used for fault tolerance but not Byzantine fault tolerant.



fig; passive model

• Active Replication;

- All replica managers are active state machines.
- Clients multicast requests to all replicas using a totally ordered, reliable multicast.
- All replicas process the same request independently in the same order.
- Tolerates crash failures and is transparent to the client.



fig; active model

Passive Replication	Active Replication
1) Only primary executes.	All replicas executes.
2) Coordination is low.	Coordination is high.
3) Fault tolerance is crash failures only.	Fault tolerance is crash & Byzantine failures.
4) Lower overhead.	Higher overhead.

Q) How does replication provide fault tolerance and high availability services to the user?

→ Fault Tolerance:

- If one replica fails (crash, network issue), other replicas continue to serve requests.
- Systems like Google File System (GFS) and Amazon S3 use replication to survive failures.

High Availability:

- Replication ensures that services are not interrupted even during:
 - Hardware failures.
 - Network partitioning.
 - Scheduled maintenance.
- Examples: DNS servers are replicated globally; if one fails, others respond.

Mechanisms:

- Failover: Automatic switch to backup replica if the primary fails.
- Heartbeat and Monitoring: To detect RM failures and redirect traffic.
- Data Synchronization: Ensures replicas stay up-to-date for reliability.